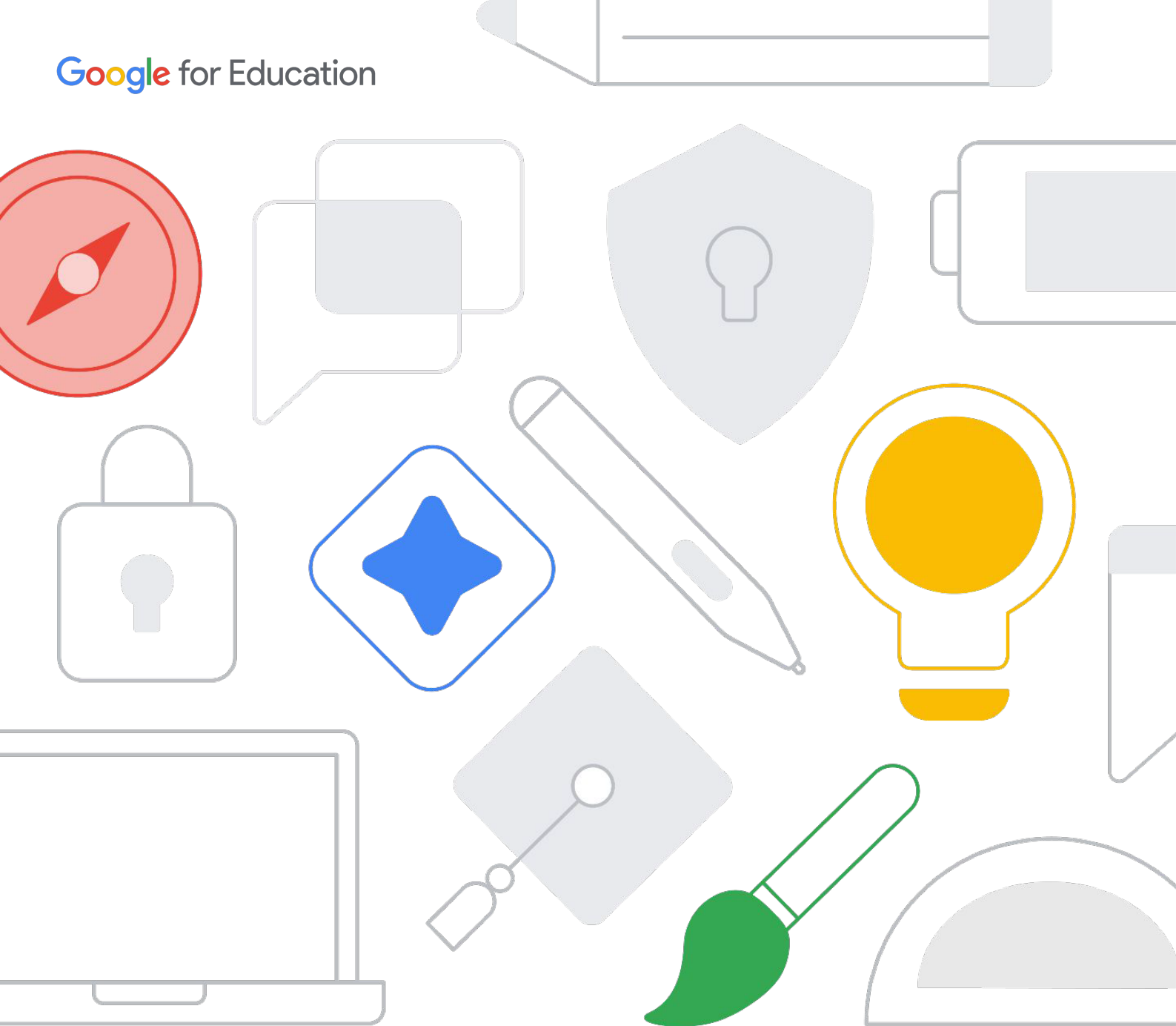




Teaching responsible use of AI

Lessons and activities for students

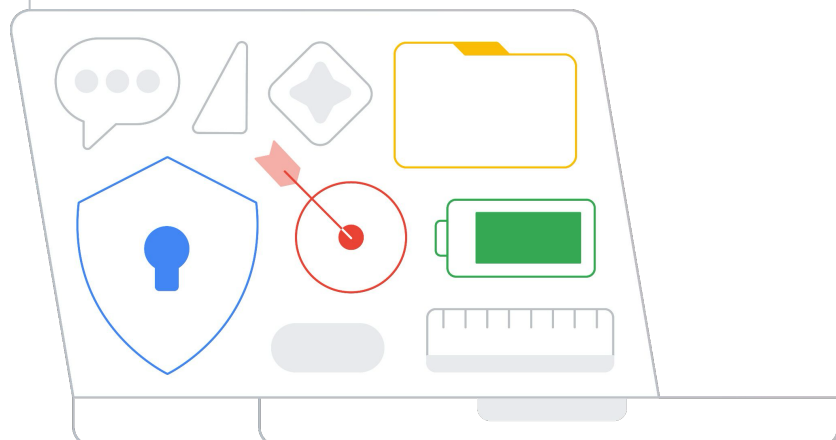
Artificial intelligence (AI) is transforming the way we live, work, and interact with technology. However, using AI responsibly requires thoughtful choices and ethical awareness. This lesson plan is designed to help teachers ensure their students become informed and responsible users of AI, like Gemini, both in and out of the classroom.

By exploring scenarios relevant to their lives, students will investigate ways AI can support them and enhance their learning, as well as potential risks and ethical implications of using AI. Students will develop strategies to navigate these challenges through guided discussions and create a personalized plan for how they will use AI responsibly.

Learning objectives

By the end of this lesson, students will be able to:

- ✓ Describe what AI is and how it works
- ✓ Describe what generative AI is and how it works
- ✓ Identify potential risks associated with using AI
- ✓ Critically examine AI outputs for accuracy and bias
- ✓ Protect their privacy while using AI
- ✓ Explain the importance of academic integrity with AI
- ✓ Discuss elements of your school's AI policy (if available)
- ✓ Differentiate between AI capabilities and human intelligence
- ✓ Identify ways to use AI to enhance, not replace, their own work in and out of the classroom
- ✓ Create a plan for making Gemini helpful in their own world, while setting ethical principles, limiting potential risks, and establishing strategies for evaluating AI-generated content



Exploring AI topics

45–60 min

Aim: Provide students with an introduction to AI and generative AI, so they can start to think critically about different scenarios involving AI use.

Things to note:

- Each scenario is discussion-based to facilitate student engagement. For each discussion, you can decide if it should engage students in small groups, in pairs, or as a class.
- This part of the lesson equips students with the knowledge they need to complete the in-class activity.

Overview of generative AI



Tip: Remember AI is technology, not human.

Explain

AI is a type of technology that identifies patterns in data to make predictions, take actions, or create content.

For example, some meteorologists use AI to help them make weather predictions.

By analyzing past weather patterns and running simulations, AI can predict future weather conditions.

- Meteorologists can use these predictions to update a weather website or app with today's forecast.
- You might have checked a weather app this morning to help you decide if you needed to grab sunglasses or an umbrella on your way to school.

Explain

Gemini is a generative AI platform. This means it can generate new content, like text, as well as music, audio files, and videos.

It has the ability to generate new content because developers train the AI model on a lot of data, or information.

- This data provides examples that the model can identify patterns in.
- Once the AI model has been trained, developers incorporate that model into AI, like Gemini, for you to use.

To use AI, you first need to type a question, called a prompt, like "What is trigonometry used for?" or "Explain the process of photosynthesis in detail" – Gemini generates a response, also called an output, based on patterns in all of the information it has access to.

- Depending on the AI used, this output might be text, images, podcasts, audio, or video.



Share a brief overview of what AI is and an example of how students might use it when they check a weather app.



Provide a brief overview of what generative AI is and how it works.

Try it on Gemini OPTIONAL

Students can try prompting Gemini with questions like, "What is trigonometry used for?" or "How does Gemini work?"

Depending on your classroom subject, you can change the example prompts. For English class, an example could be: "Explain the difference between active and passive voice."

Explain

And as our world constantly changes, Gemini evolves too by improving pattern recognition and the ability to interpret and produce diverse types of information.

This means that the response, or output, you may have gotten from Gemini today may be very different from what you would get two years from now.

The output generative AI models give you at any point in time are solely based on how they are trained.

All AI models, like Gemini, lack consciousness, the ability to think independently, or have and understand emotions.

- If you ask Gemini where you should go to college, or how you should spend your earnings from your summer job – Gemini can provide ideas, but the decisions are yours to make, so you don't need to follow its advice. You also should never use Gemini for decisions relating to your personal, health or safety – you should consult an adult who you trust.

Key takeaways

With this in mind, the potential of AI is vast. Have fun exploring the different ways AI can support you.

For example:

- If you are trying to understand a complex concept, you can use Gemini to summarize it or explain it in simpler terms.
- If you have some great ideas for a presentation, but need some help with structuring them, Gemini can provide suggestions on how to organize them.
- It can also explain grammar and punctuation rules to help you figure out how to avoid comma splices when you write.

Remember: AI is technology, not human.

So, even though its capabilities are exciting, AI doesn't have human intelligence. It can't think for itself, or feel emotions; it's just great at picking up patterns, known as training. Since AI is not a human, it can't and shouldn't make decisions for you or replace important people in your life.



Emphasize how AI is not the same as human intelligence.

 Try it on Gemini OPTIONAL

Have students ask Gemini a more personal question, like where should you go to college?

This will help show that it's important to remember that AI can help students think about these types of questions, but students shouldn't do exactly what it tells you.



If you are interested in diving deeper with students into how AI works and the history of AI, you can check out these additional resources:

[Experience AI](#)

[Explore a Topic: Generative AI](#)

[Discover AI in Daily Life](#)

Examining output for accuracy and bias



Tip: Always critically evaluate responses.

Tip: If something feels off, investigate further.

Explain: Hallucinations

Now that we've discussed some of the helpful ways you can use AI, like Gemini, let's explore how to examine an output to make sure it's correct.

We can start by thinking about how you develop knowledge.

Your knowledge comes from a wide variety of sources: Your family, friends, classmates, teachers, experiences, experiments, videos, music, books, etc.

When you're asked a question, you use all of your knowledge to come up with the best answer.

When we as humans misremember information, we sometimes make things up to fill in the gaps, even if it's wrong.

- Like, have you ever gotten a question wrong on a test?
- Or have you ever tried to explain a process in an essay, like a plant's life cycle, but you forgot some of the steps?

Because the AI is always evolving, it can sometimes give you incorrect responses. When AI invents an answer, it's called a hallucination.

Hallucinations can happen when you ask a question about really specific information that AI doesn't have access to yet, so the answer it gives is made up.

AI can also generate responses that are not relevant to the question you asked or might miss key points and details.

If AI doesn't understand the question, it might give you an incorrect answer.

- **For example:** If you ask an unclear question about bats (the animal), the AI might misinterpret the language and give you information about sports bats instead.

Generative AI also has limitations with the ability to reason or solve problems. Most of the time, AI can only solve problems correctly if it has been trained on similar examples. If not, it may generate more hallucinations. With all these limitations in mind, it is very important to double check the accuracy of the information you generate with AI in other places like Google Search.



Discuss how we develop knowledge and how we sometimes misremember information and come up with an incorrect answer.



Discuss how AI can give incorrect answers. And define hallucination.



Discuss the different limitations of AI and the outputs they give you (For example: Responses that are irrelevant to the prompt or are missing information).



If time permits, you can show students a [video](#) of how the fact-checking feature works when they use Gemini.

✦ Try it on Gemini

OPTIONAL

You can also open Gemini and type in a prompt to show students in real-time how the fact-checking feature works.

Ask

Now, think about a time a friend or classmate might have asked you for a recommendation, like a new music playlist or music channel to stream.

How did you come up with your recommendation?

Most likely, it was based on your listening preferences. And your preferences were probably shaped by what your family members listen to, what your friends listen to, and what music genres you've been exposed to.

With that in mind, is your music recommendation biased?

If you're not sure, let's quickly review what a bias is.

A bias is a prejudice in favor of, like in this situation, or against one thing, person, or group compared with another.

If your music recommendation is based on your preferences, then is it biased?

- Discuss their responses.
- Explain that an answer based on personal preferences, in which you favor one thing over another, is biased.

Let's consider another example. If a classmate asked you which high school has the best basketball team, will your answer be biased?

- Discuss their responses.

People, in general, have biases. We have favorite foods, favorite activities, favorite movies, favorite sports teams, etc.

People train AI models, like Gemini, and people put information on the internet that Gemini can access. All these pieces of information have the potential to be biased towards a particular way of thinking.

Consequently, when you type something into Gemini, the response you get may be biased or may contain biased content.



Introduce the concept of bias to help students understand that AI can provide responses that are biased.



Give your students a minute to consider their answer. If necessary, review the meaning of bias.



Give your students a minute to consider their answer.



Give your students a minute to consider their answer.



If time permits, you can show students a [video](#) about bias and how industry experts combat bias.

Discuss

So now we know Gemini can give you wrong answers, answers with missing information, and answers that are biased.

With this in mind, ask students to think about the following scenario:

Leo wants to study entomology and is working on a research paper about a beetle species that he believes to be the rarest insect. He is struggling to find information to validate his assumption. So he decides to use Gemini, hoping it will provide him with the information he needs. Leo types in his question, and Gemini provides him with a response that describes the beetle as the rarest insect. Leo includes this information in his essay.

- What is your initial reaction to this scenario? Is a species of beetle actually the rarest insect?
- What can Leo do to check if the information Gemini provided is correct?
- How can Leo figure out if the information Gemini provided is biased?

Explain: Deepfakes

Another topic related to inaccurate information that students should be aware of when working online are **deepfakes**.

Have students think about the following scenario:

What if you read a headline on social media that said: "World's Richest Person Buys the Moon – and Plans to Build a Theme Park!"

- What are your first impressions?
- Does this seem unusual?
- What would you do if you saw this online?

If the content seems unusual, pause and investigate further.

Like every useful technology, there may be people who try to take advantage of AI to deceive or defraud others.

- For example, they may generate misinformation or fake media, like photos that seem real.

Remind students that creating and distributing deepfakes is illegal.

A deepfake is a digitally-manipulated video, photo, or audio recording created using someone's voice and/or likeness without their permission.

Always check out the content's sources and consider the intent of who published it.



Recommend discussing in small groups or in pairs, but the discussion can be the entire class.



Discuss how we develop knowledge and how we sometimes misremember information and come up with an incorrect answer.



If you want to dive deeper into misinformation, deepfakes, and fraud with your students, check out the resources on [YouTube's Hit Pause](#) program.

Key takeaways

Gemini is a great resource if you ever get stuck researching a topic.

Just like how Leo couldn't find information on if the species of beetle is the rarest insect, you can ask Gemini a question about your topic to find out if that information exists.

Or if your topic is unfamiliar to you, using Gemini can be a helpful starting point.

Then, instead of copy and pasting the output Gemini gives you into your assignment, you can use the language or information from the output to perform additional research.

This approach will help you check if the information Gemini provides is correct and unbiased.

Gemini also has a fact-checking feature, which is a great resource for verifying if an output is accurate. Once you learn that an output is correct, you need to find the source of the information that Gemini provides to reference it in our assignment.

Also, below the search bar in Gemini, it states: "Gemini can make mistakes, so double-check it," which is a good reminder why we don't assume outputs are always correct.

Remember, if something seems unusual, pause and investigate further.

Remember: Critically evaluate responses.

Since generative AI is a work in progress, it can make mistakes and may even make things up – known as hallucination.

Always check information that's presented as fact.

When in doubt, double check the response.

Remember: If something feels off, investigate further.

Like every useful technology, there may be people who try to take advantage of AI to deceive or defraud others. If the content seems unusual, pause and investigate further.

Check out the content's sources and consider the intent of who published it.



Protecting privacy



Tip: Keep private information private.



Explain

Let's explore another scenario:

You're playing an online game and you make a new friend.

She suggests that you exchange numbers.

Later on, you get a text from her. She asks you where you were born and your birth date. And then she asks you about your family members' names.

Her requests become more personal, inquiring about your different email addresses and if you have a bank account.



Discuss a scenario about keeping private information private.

Discuss

How do you respond to her texts?

- Give your students a minute to consider their answer.
- Discuss their responses.

Remember, you do not know this person other than your online interactions while you play a video game.

- Why is she asking you for all of this personal information?
- What is she going to do with all of this personal information?

Let the students discuss.

Now, if you do not feel comfortable sharing all of your personal information with someone you don't know, should you share it with Gemini?

- Discuss their responses.



Recommend discussing in small groups or in pairs, but the discussion can be the entire class.

Explain

For many of us, when a person we do not know asks for our personal information, we become cautious, as the risks of providing private information are clearer.

With technology, especially AI, the risks are less clear. But we still need to be just as cautious.

There is a wide variety of AI available for you to use. But they are not all created the same.

For example, some AI may use the information you input into it, including private information, to train its model.

Fortunately, Gemini is designed with privacy in mind.

When you use Gemini as a young person, there are protections in place to keep your information private:

- The activity recording is off, so Gemini won't keep a record of what you input.
- Filters block any private information included in a prompt, like your birth date, from being reviewed or used to train Gemini.
- That being said, you still shouldn't share your personal information with any AI model.

All AI is created differently, meaning you should take the time to review and understand their privacy practices.



If time permits, consider showing the students the Google for Education [Guide for AI](#) and exploring some of the privacy practices with them.

Key takeaways

Even though there are privacy protections when you use Gemini as a young person, it's important to limit risks and build good habits by keeping personal information private.

Here are some examples of private information to not share with AI:

- Full name, social security number, birth date, home address, phone number, email address, driver's license number, passport number, any financial or health information.
- These include text and images – please do not upload an image of your driver's license, for example.



Remember: Keep private information private.

Never enter personally identifiable information, such as your social security number, into generative AI systems. When in doubt, consider whether you would share this information publicly.



Academic integrity



Tip: Use AI to boost your talents, not replace them.



Discuss

Let's explore another scenario:

Your classmate, Ahmed, is behind with writing his art history report. It is due during finals week, and he struggled managing his time to prepare for all of his exams and finish his projects. He knows that Soraya excels in the class, so he asks if she can write the report for him. Since Soraya has completed her work already, she agrees to help and write the rest of Ahmed's paper.

- What is your initial reaction to this scenario?
- What is problematic about this scenario?



Recommend discussing in small groups or in pairs, but the discussion can be the entire class.



Students should identify that Ahmed cheated: He used another person's work as his own for a grade.

Explain

Ahmed's actions are a breach of academic integrity.

Academic integrity means doing your work honestly in a way that is fair to you and others.

Different schools and teachers have their own academic integrity policies, so it is important to know them and understand the guidelines you need to follow.



Discuss

Now, instead of asking Soraya to finish the report, Ahmed types the missing content he needs into Gemini and copies and pastes the outputs into his paper. Then, he submits the report for a grade.

- What is your initial reaction to this scenario?
- What could Ahmed have done differently in this scenario?
- Do you think copying the output directly from Gemini and submitting it is cheating?

What could Ahmed do instead?

- Ahmed could write a prompt, asking Gemini for ideas, to help brainstorm different ways to approach finishing the paper.
- Gemini can be great for brainstorming. Brainstorming sessions are a valuable practice for coming up with ideas or overcoming writer's block. You can ask questions, ask if you are missing some information, or ask to elaborate on an idea or topic.
- He could ask Gemini for creative ideas for concluding an art history report to inspire his approach for writing a fun and engaging conclusion.
- Or if he needs more information, he could ask Gemini questions about his topic, and use the language from the outputs or sources listed in the outputs to research and cite credible sources.

Key takeaways

Ahmed copying content from Gemini and using it as his own is the same as asking Soraya to finish writing the paper for him.

Ahmed is taking credit for work that is not his.

Gemini can help Ahmed finish the paper without copying.



Remember: Use AI to boost your talents, not replace them.

AI can help you kickstart the creative process, but it's not there to do the work for you – that's your role as the creator.



This is an opportunity to show the ways Gemini can be helpful vs. using Gemini outputs as their own work.

Try it on Gemini

OPTIONAL

If time permits, you can open Gemini and show students how they can use it to brainstorm ideas for their next project or subject.

For ideas on what to brainstorm, check out this guide on [learning in creative ways with gen AI](#).



For additional resources on evaluating credibility of online sources, you can check out:

- An Applied Digital skills lesson: [Evaluate Credibility of Online Sources](#)
- [Hit Pause](#) from YouTube







AI policies in school

Explain

In addition to academic integrity policies, schools and teachers also have AI use policies and guidelines.

These policies vary by school and sometimes by teacher, so it is important to understand the AI guidelines your school and teachers expect you to follow.

The purpose of an AI policy is to give you guidance on the best ways to use AI, while also ensuring everyone's work remains honest and fair.

Policies play a very big role in ethical use of AI.

- When you do something ethically, you do what's right. Ethics are like your moral compass—pointing you towards fair, honest, and kind actions.

These policies are designed to make sure you are learning. They support critical thinking by limiting your reliance on AI. Critical thinking helps you solve problems, make good decisions, and understand the world around you. Without it, how can you tell the difference between a fact and a made-up story? Or figure out a solution to a complex problem?

Key takeaways

If it helps, you can think of AI policies and guidelines like a baking recipe.

- All of the ingredients and steps are listed for you to follow.
- If you follow the baking recipe and put it in the oven, your result will likely be positive—a delicious dish.
- But if you don't follow all of the steps or skip ingredients, your result will probably be disappointing.

Follow your school's AI policies, so we enjoy the benefits of AI while using it in positive, ethical, responsible, and productive ways.



If you have one, use this opportunity to share your school's (and your own) AI guidelines with the class.

In-class activity

30-45 min

Aim: Get students to consider everything they learned about ethical and responsible AI practices and use them as a guide for creating a list of ways they could use AI in school to support and enhance their learning.

Once they come up with their list, they will make a creative representation about their learnings to share with their classmates. If time permits, ask students to present to the class.



Activity steps

Explain: Overview

With all this new information fresh in your mind, we are going to create a plan for responsible AI use. But this is no ordinary plan. The challenge is to make it as attention grabbing as possible. This could be in any format you can think of – a personal AI charter, a poem, a poster for your grandparent, a recipe, or even a movie script. The sky's the limit!

The goal? Anyone in this school should be able to read your plan and know how to use AI responsibly.

Step 1: Make the list

In a Google doc (or similar program) or in your notebook, list how you could use AI platforms, like Gemini, while you're in school.

Think about the ways Gemini, and similar platforms, can help enhance your learning and give an example. Here are some to get you started:

- AI could help you figure out how to approach writing an email to ask a teacher for a letter of recommendation.
- Or it could help with time management by creating homework and study schedules.
- Gemini could help you brainstorm career paths and plan for your future. You could research vocational & trade schools, entrepreneurship, online courses/certificates, gap year, and colleges.

Then, think about what you should be watching out for when using Gemini or other AI platforms, leaning on the 5 AI tips for reference:

- 1 Remember AI is technology, not human.
- 2 Critically evaluate responses.
- 3 If something feels off, investigate further.
- 4 Keep private information private.
- 5 Use AI to boost your talents, not replace them.

Step 2: Create the plan

Next is the fun part – decide how you want to present your list. Get as creative as possible.

- Add in images, color, fun metaphors, drawings – whatever you can think of to bring your list to life in the most attention grabbing way.
- **For example:** Fun format idea – use items from your list to create song lyrics or a poem.

Try it on Gemini

You could ask Gemini to think of some ideas for song lyrics or a poem based on your list.

- Using either the [in-class activity template](#), the [print student handout](#) or a physical poster board, the aim is to capture the attention of your classmates.



You can show students an example in the slide deck.



You can use the [in-class activity template](#) to share with the class.

Students all work in the same deck, each developing their own slides (recommended). Or you can ask students to create their own deck and share the link with you and their classmates.



You could also suggest Google Drawings or Sheets as other tools students could use.

Step 3: Share with the class

Depending on time, ask for a few volunteers to share their creative AI use plan with the class.

If students needed most of the allocated time to develop their creative AI plan, you could instead:

- Ask students to put all their work in the shared [in-class activity template](#) deck and view their classmates' plans.
- Ask students to post/hang their poster or page from the [print student handout](#) up in the classroom or at their desk for others to view throughout the day.

After the lesson

Reflection

10 min

Aim: To inspire students to reflect on what they have learned and consider how this knowledge will influence how they use AI in the future.

Note: We have two additional online lessons on AI if students are interested in exploring the topic further.

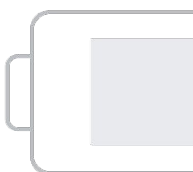
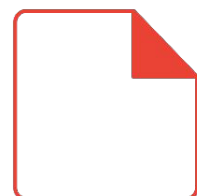
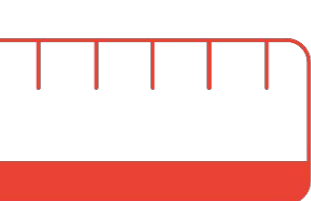
Discuss

- How will your AI use plan be helpful in school?
- What did you find most interesting about creating your plan?
- What did you find most challenging about creating your plan?
- Reflecting on your AI use before this lesson and what you learned today, how do you feel about using AI, like Gemini, in school?
- How have your thoughts and opinions about using AI changed?
- What is your biggest takeaway from this lesson? And please explain why.

Explain

If you are interested in learning more about AI and different ways it can support you in school, here are some additional online lessons to explore:

- [Explore a Topic: Generative AI](#)
- [Discover AI in Daily Life](#)



Customize your lesson



Short on time? Try these options:

- Break the lesson up over two class periods.
- Assign all or part of the in-class activity as a homework assignment.
- Only ask students to create lists and skip the visual portion of the in-class activity.
- Use the next class period for presentations on the AI use plan.
- Assign the reflection questions as homework.



Need examples and scenarios tailored to your class? Try these options:

After reviewing the examples and scenarios in the lesson plan, consider ways you can modify the content to align with what you're teaching.

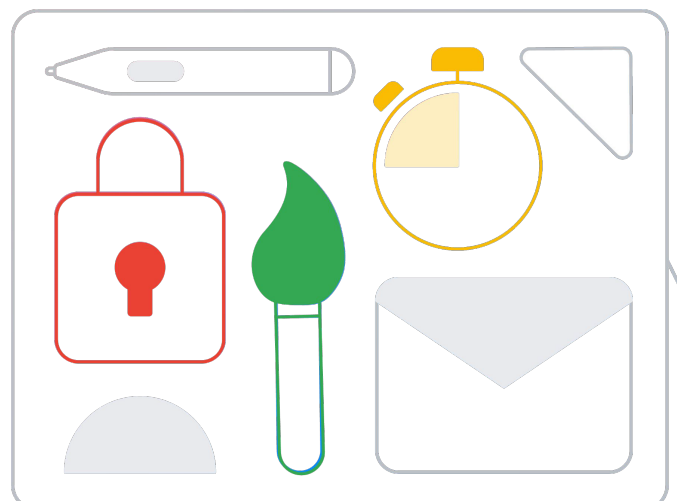
- For example, in Class discussion 1 and 3, instead of a research report on a beetle or an art history essay, you can substitute these assignments with ones that you assigned or will assign in your class.

You can also ask Gemini to take the scenarios and make them relevant to your class and what you are learning, or ask it to generate alternative scenarios all together.



Have extra time? Try these options:

- Assign additional steps for the in-class activity.
- Encourage students to dedicate more time to the layout, style, and/or visual elements of their deck or poster board.
- Have every student present their creative AI use plan to the class (or in small groups).
- Ask for volunteers to present their creative AI use plan to the class.
- Explore in more depth how AI works and the history of AI by showing the class the video: [Experience AI](#).
- Teach an AI lesson from the Applied Digital Skills curriculum: [Explore a Topic: Generative AI](#), [Discover AI in Daily Life](#).
- Teach a lesson about evaluating the credibility of online sources from the Applied Digital Skills curriculum: [Evaluate Credibility of Online Sources](#).



Customize your lesson



Need learning support?

Try these options available in Chromebooks and Google Workspace.

- For motor and dexterity support: Go keyboardless and use **voice typing** to enter, edit, and format text in Google Docs and Slides.
- For display and vision support: Make reading easier with **reading mode**, which can read text aloud, summarize information, highlight specific lines, and more. Reduce eye strain and increase focus with **dark mode** on Google Classroom app.
- For cognitive and spoken feedback support: Help students focus with Google Meet pinning, which allows them to view just the teacher and hide other participants.
- For audio support for remote learners: Turn on **captions** in Google Meet to view subtitles in real time.
- For reading support: Make billions of PDFs accessible to screen readers with built-in **PDF Ocular Character Recognition (OCR)**.
- Check out our [YouTube playlist](#) on accessibility features available across Chromebooks and Google Workspace for Education to learn more.

We'd love your feedback to continue improving this teaching resource.
[Please fill out this survey](#) to share your thoughts.

